

II. THEORETICAL FRAMEWORK

The starting point of the DDRHF theory is an effective Lagrangian density $\mathcal{L}$ [21], which contains the degrees of freedom associated with the nucleon field (ψ), two isoscalar meson fields (σ and ω), two isovector meson fields (π and ρ) and the photon field (A). Then the effective Hamiltonian $\mathcal{H}$ is obtained with the general Legendre transformation. On the level of the mean field approximation, the energy functional $\mathcal{E}$ is obtained by taking the expectation of the Hamiltonian $\mathcal{H}$, where both the Hartree (direct) and Fock (exchange) terms are kept. Finally, the Dirac equations, i.e. the equations of motion of nucleons, are obtained via the variation of the energy functional $\mathcal{E}$.

For spherical nuclei, the nucleon Dirac spinor can be written as,

$$f_\alpha(\mathbf{r}) = \frac{1}{r} \begin{pmatrix} iG_{n_a}(r)\mathcal{Y}_{j_a m_a}^{l_a}(\hat{\mathbf{r}}) \\ -F_{\tilde{n}_a}(r)\mathcal{Y}_{j_a m_a}^{\tilde{l}_a}(\hat{\mathbf{r}}) \end{pmatrix} \chi_{\frac{1}{2}}(\tau_a), \quad (1)$$

where $\chi_{\frac{1}{2}}(\tau_a)$ is the isospinor, $\mathcal{Y}_{j_a m_a}^{l_a}$ is the spherical harmonics spinor and $\mathcal{Y}_{j_a m_a}^{\tilde{l}_a}(\hat{\mathbf{r}}) = -\hat{\boldsymbol{\sigma}} \cdot \hat{\mathbf{r}} \mathcal{Y}_{j_a m_a}^{l_a}(\hat{\mathbf{r}})$ with $\tilde{l}_a = 2j_a - l_a$. For the negative energy states, the lower component $F(r)$ is dominant. The states are labelled by $\{\tilde{n}\tilde{l}jm\}$ with the relation

$$n = \tilde{n}, \quad \text{for } \kappa > 0, \quad n = \tilde{n} + 1, \quad \text{for } \kappa < 0, \quad (2)$$

in analogy to Ref. [26]. The spin symmetry concerns the near degeneracy of the states $(\tilde{n}, \tilde{l}, j = \tilde{l} \pm 1/2)$. In the following equations, the sub-index will be omitted for simplicity.

The radial Dirac equations are the coupled integro-differential ones due to the non-local Fock terms X and Y [18],

$$EG(r) = - \left[\frac{d}{dr} - \frac{\kappa}{r} \right] F(r) + [M + \Sigma_S(r) + \Sigma_0(r)] G(r) + Y(r), \quad (3a)$$

$$EF(r) = + \left[\frac{d}{dr} + \frac{\kappa}{r} \right] G(r) - [M + \Sigma_S(r) - \Sigma_0(r)] F(r) + X(r). \quad (3b)$$

Introducing the effective local potentials X_G, X_F, Y_G and Y_F by the definitions,

$$X(r) = \frac{G(r)X(r)}{G^2 + F^2} G(r) + \frac{F(r)X(r)}{G^2 + F^2} F(r) \equiv X_G(r)G(r) + X_F(r)F(r), \quad (4a)$$

$$Y(r) = \frac{G(r)Y(r)}{G^2 + F^2} G(r) + \frac{F(r)Y(r)}{G^2 + F^2} F(r) \equiv Y_G(r)G(r) + Y_F(r)F(r), \quad (4b)$$